

Jatin Shukla

Noida, Uttar Pradesh — +91-9116237146 — shukla.jeetu2550@gmail.com — [LinkedIn](#) — [GitHub](#) — [Portfolio](#)

Professional Summary

Senior AI/ML Engineer with **around 3 years of total experience** building production-grade AI agents, RAG systems, machine learning models, and full-stack AI applications. Strong expertise in Python, SQL, FastAPI, PyTorch, Scikit-learn, LangChain, semantic retrieval, prompt engineering, vector search, and cloud deployment. Proven ability to translate business problems into scalable AI solutions, improve model quality and latency, and deliver measurable impact through user-facing AI products and intelligent automation.

Technical Skills

Languages: Python, SQL, JavaScript, C++, C#

AI / ML: Machine Learning, Deep Learning, PyTorch, Scikit-learn, Regression, Classification, Clustering, Feature Engineering, Model Evaluation, EDA, Experimentation

GenAI / Agents: AI Agents, LLM Applications, RAG, Prompt Engineering, LangChain, Semantic Search, Context Grounding, Information Retrieval, Agentic Workflows

Data / Retrieval: FAISS, Vector Search, BM25, Cross-Encoder Reranking, Document Parsing, Unstructured Data Processing, TF-IDF, Cosine Similarity, Sentence Transformers, Resume & JD Parsing

Frameworks / Deployment: FastAPI, Flask, React, Vite, Tailwind CSS, REST APIs, Docker, Azure, CI/CD, Vercel, Render, Joblib, Git, GitHub

Experience

Senior Engineer – AI/ML

Mar 2026 – Present

Havells India Ltd., Noida

- Built AI agents for applicant shortlisting for HR workflows, automating resume screening, JD matching, and ranking logic to reduce manual review effort by **40%+** and accelerate shortlist turnaround.
- Developed agentic ideation, research, and prototyping workflows for internal use, enabling structured **10-step reasoning pipelines** for faster concept exploration and improving early-stage solution turnaround by **30%**.
- Enhanced internal AI products with retrieval, prompt orchestration, scoring logic, and user-facing workflows, improving usability and consistency for business teams across model-assisted decision tasks.

AI/ML Engineer

Jan 2025 – Feb 2026

Havells India Ltd., Noida

- Built and deployed ML models for the fan team to predict air delivery, velocity, and pressure on fan blades, improving engineering prediction accuracy by **18–22%** and reducing iterative validation effort.
- Developed ML models for the LED team to estimate heat generation at critical points, supporting thermal-performance analysis and reducing design evaluation turnaround by **35%**.
- Engineered a production inference pipeline for a Tri-Branch PointNet regression model, reducing end-to-end latency by **35%** through optimized preprocessing, feature scaling, and joblib-based deployment.
- Designed automated data preprocessing, feature engineering, and inference consistency pipelines using Python, PyTorch, Scikit-learn, and joblib, improving model reliability across environments.
- Built UI-driven internal applications for model interaction using Flask, React, and VTK.js, enabling **30+ engineers** to use AI outputs through business-friendly interfaces and cross-platform packages.
- Created scalable 3D / point-cloud evaluation workflows handling **10+ design configurations per week**, reducing manual analysis and dependence on physical prototyping.

Projects

Nexora – AI Study Assistant

Personal Project — [Live](#) — [GitHub](#)

- Built Nexora, a full-stack RAG-based AI knowledge assistant using React, Vite, FastAPI, and Supabase to manage app data and support grounded Q&A over PDFs and web URLs.
- Engineered a hybrid retrieval pipeline with FAISS for dense semantic search, BM25 for sparse retrieval, and Cross-Encoder reranking, improving relevance over vector-only search.
- Developed an AI learning workflow with quiz generation, scoring, explanations, weak-topic feedback, and 6 answer modes, turning static content into an interactive study system.

HireFit – AI Resume–Job Matcher

Personal Project — [Live](#) — [GitHub](#)

- Built a full-stack AI/ML and GenAI application using FastAPI, React, Vite, Tailwind CSS, Python, and Gemini API for resume-JD matching, ATS scoring, semantic comparison, and recruiter-facing recommendations.
- Designed structured evaluation pipelines combining TF-IDF, cosine similarity, evidence validation, prompt-based reasoning, and multi-factor scoring for accurate profile-to-role assessment.
- Implemented features including keyword coverage analysis, shortlist-risk analysis, safer resume rewrites, multi-JD comparison, and export-ready outputs for end users.

AgentFlow – Multi-Agent AI Platform

Personal Project — [Live](#) — [GitHub](#)

- Built a full-stack multi-agent AI orchestration platform using FastAPI, LangGraph, Groq, React, and SQLite, routing tasks across 4 specialist agents for research, coding, writing, and analysis.
- Designed a 6-stage controlled AI workflow with memory retrieval, supervisor planning, backend tools, reviewer scoring, and human-in-the-loop approval to improve reliability.
- Implemented workspace-isolated memory, run history, and trace persistence capturing agent decisions, tool usage, review scores, human actions, and final outputs.

IoT-Enabled Digital Twin for Post-Harvest Loss Minimization

Research Project — [Link](#)

- Built an Azure-based AI + IoT system using Raspberry Pi and sensors for predictive analytics, real-time monitoring, and decision support, reducing potential post-harvest losses by **15–20%**.

Education

National Institute of Technology, Hamirpur

M.Tech, Computer Science and Engineering

2023 – 2025

CGPA: 9.15/10

Institute of Engineering and Technology, Lucknow

B.Tech, Chemical Engineering

2018 – 2022

CGPA: 7.8/10